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SYSTEM, APPARATUS AND METHOD FOR ACCESSING MULTIPLE ADDRESS SPACES VIA A VIRTUALIZATION DEVICE

Abstract

In one embodiment, an apparatus includes an input/output virtualization (IOV) device comprising: at least one function circuit to be shared by a plurality of virtual machines (VMs); and a plurality of assignable device interfaces (ADIs) coupled to the at least one function circuit, wherein each of the plurality of ADIs is to be associated with one of the plurality of VMs and comprises a first process address space identifier (PASID) field to store a first PASID to identify a descriptor queue stored in a host address space and a second PASID field to store a second PASID to identify a data buffer located in a VM address space. Other embodiments are described and claimed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of application Ser. No. 16/909,068, filed Jun. 23, 2020, which Indian Provisional Patent Application No. 202041018444, filed on Apr. 30, 2020, in the names of Sanjay Kumar; Philip Lantz; Rajesh Sankaran; Narayan Ranganathan; Saurabh Gayen; David Koufaty; and Utkarsh Y. Kakaiya, entitled “SYSTEM, APPARATUS AND METHOD FOR ACCESSING MULTIPLE ADDRESS SPACES VIA A DATA MOVER,” the disclosures of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] Embodiments relate to data operations in a computing system.

BACKGROUND ART

[0003] A data mover is a device/accelerator that offloads memory copy and other related operations from a processor. For example, besides doing memory copy from source to destination, a data mover can also do a variety of other operations on memory. Data movers may support shared virtual memory (SVM), where they can access the same virtual address space as user space applications. However, state of the art data mover devices supports only a single address space per job descriptor, which limits use cases.

[0004] Many computing systems also provide support for scalable input/output virtualization (IOV) devices. These devices, which may include network interface circuits and memories among others, define assignable device interfaces (ADIs) that can be assigned to a virtual machine (VM) as part of a virtual device (VDEV) emulation. The ADIs are configured with a process address space identifier (PASID) to access the VM's memory directly. However, using only a single PASID on the ADIs may cause high overhead in certain scalable IOV device usages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The invention may best be understood by referring to the following description and accompanying drawings that are used to illustrate embodiments of the invention. In the drawings:

[0006] FIG. 1 is a block diagram of a data mover in accordance with an embodiment of the present invention.

[0007] FIG. 2 is a block diagram of a memory move job descriptor in accordance with an embodiment.

[0008] FIG. 3 is a block diagram of a PASID permissions table according to one embodiment.

[0009] FIG. 4 is a flow diagram of a method in accordance with one embodiment.

[0010] FIG. 5 is a flow diagram of a method in accordance with another embodiment.

[0011] FIG. 6 is a flow diagram of a method in accordance with yet another embodiment.

[0012] FIG. 7 is a block diagram of a computing system in accordance with an embodiment of the present invention.

[0013] FIG. 8 is a block diagram of a system in accordance with another embodiment.

[0014] FIG. 9 is a flow diagram of a method in accordance with yet another embodiment of the present invention.

DETAILED DESCRIPTION

[0015] In various embodiments, offload devices in various computing systems may be configured in a manner to enable such offload devices to perform data operations amongst multiple address spaces. With this arrangement, various memory-based tasks can be offloaded to these offload devices. Still further, such operations may be performed more efficiently by enabling a single device, in response to a single request, to access securely multiple address spaces (or at least portions thereof). As a result, various overheads in obtaining precursor data for such offload operations from different address spaces and collecting them into a single address space can be avoided. While embodiments herein are described in the context of data mover devices and virtualization devices such as input/output virtualization devices, understand that embodiments are not limited in this regard, and the techniques described herein can be used in a wide variety of computing environments.

[0016] In contrast, typical data movers operate on a job descriptor that is associated with a single process address space ID (PASID) corresponding to the application that submitted the descriptor. As part of descriptor processing, the data mover may access the submitter's memory (for source, destination, and completion addresses) using the PASID of the descriptor. However, using only a single address space for processing a descriptor limits various use cases where it is desired to access multiple address spaces. With embodiments herein, a process having a PASID (e.g., PASID X) may submit to a data mover a job descriptor having additional information to enable access to address spaces in addition to the process's own address space.

[0017] Although embodiments are not limited in this regard, various use cases may benefit from submitting a job descriptor having information for multiple PASIDs to enable a data mover to securely access multiple address spaces in response to this single job descriptor. One example is in the context of a virtual switch (vswitch) in a virtual machine monitor (VMM). This vswitch provides the functionality of switching packets between virtual machines (VMs) on the same system. As an example, the vswitch may be used to directly copy packets from a source VM's memory to a destination VM's memory. With embodiments, a job descriptor may provide information including addresses from three different address spaces, i.e., source address from source VM, destination address from destination VM, and completion address (to notify of completion) from vswitch application in the VMM. And as such, a data mover may, in response to this job descriptor, perform this memory operation to directly copy the packets, securely accessing three different address spaces.

[0018] As another example, in a multi-key total memory encryption (MKTME) usage where a different key is used to encrypt each VM's memory and the key ID is stored in cache lines of a central processing unit (CPU), a VMM may seek to flush CPU caches when it switches a VM on a CPU. With an embodiment, the VMM may offload this CPU cache flush for all of the VM's memory to the data mover by providing a job descriptor having addresses from two different address spaces, i.e., cache flush address range from the VM and completion address from the VMM. A still further example is in the context of shared memory usage. Assume here that two processes (e.g., message passing interface (MPI) ranks) may setup a shared memory and send/receive data through the shared memory pages. These processes may use a data mover for copying data between the two address spaces, via a job descriptor that includes addresses from two different address spaces (corresponding to the two applications).

[0019] Of course, a data mover may be configured to perform a variety of additional memory

operations, in addition to copy or move operations. As examples, a data mover may also be configured to perform memory fill, copy with cyclic redundancy check (CRC), memory compare, dualcast, generating memory diffs between two memory regions, applying diffs to a memory region, flush CPU caches, etc.

[0020] In addition, embodiments enable a data mover to control access to a limited address range within address spaces. With embodiments, a data mover may control access to an address space on a more fine-grained manner for better isolation. For example, for some of the above usage cases, a data mover may control access to limited address ranges within an address space.

[0021] With embodiments, a data mover may be configured to securely access multiple address spaces in response to a single job descriptor that specifies multiple PASIDs. To verify an access requested by a first agent to an address space of a second agent is permitted, the data mover may include or be coupled to a PASID permissions table (PPT). This permissions table may be used to validate a job submitter's access permissions for accessing the other PASIDs, as well as limiting access to certain address ranges. The data mover may be configured by writing to the PASID permissions table directly by the privileged data mover driver software. In addition, the data mover may be configured, in response to a setup descriptor, to allow an unprivileged client to directly setup and modify its own address ranges and access permissions in the PASID permissions table.

[0022] In this way, a data mover can access memory from multiple address spaces to process individual jobs. As a result, clients and system software can avoid: 1) copying data into a single address space; or 2) mapping data into a single address space. Embodiments thus may reduce the overhead of data mover operations and enable offload from the CPU for one or more usage cases.

[0023] Referring now to FIG. 1, shown is a block diagram of a data mover in accordance with an embodiment of the present invention. As shown in FIG. 1, data mover **100** may be a hardware accelerator configured to perform data movement operations offloaded from a CPU or other requester. As such, data mover **100** may be implemented as a hardware circuit included in a processor such as system on chip (SoC), e.g., implemented on the same semiconductor die or within a common package or socket with the SoC. In yet other cases, data mover **100** may be implemented as a standalone device such as may be implemented within a datacenter system. In many implementations, data mover **100** may appear to the CPU or another requester as a peripheral device, e.g., a Peripheral Component Interconnect Express (PCIe) device.

[0024] In any event, as shown in the high level of FIG. 1, data mover **100** includes a control circuit **110** configured to receive incoming job descriptors from one or more requesters. While described herein as receiving offload requests from a CPU or other agent, understand that more typically these requests may be initiated in response to a memory operation requested by a software agent such as a given application, virtual machine, container, virtual machine monitor or other such software agent. Such requests may thereafter be offloaded from CPU to data mover **100**.

[0025] In embodiments herein this job descriptor may include information to enable data mover **100** to access multiple address spaces. To this end, to verify that a requester is permitted to access additional address spaces, certain information included in the job descriptor may be used to access a PASID permissions table **120**. As will be described herein, permissions table **120** may include multiple entries each to store information regarding permitted accesses to a given address space (referred to herein as an access address space) by a particular requester (also referred to herein as a submitter).

[0026] Assuming permission is verified, control circuit **110** may send information associated with a requested memory operation to one of multiple execution circuits **130.sub.0-130.sub.n**. Execution circuits **130** may be pipeline circuits that execute the requested operation such as a cyclic redundancy checksum (CRC). As will be described herein, each execution circuit **130** may be configured to issue memory operations to one of multiple address spaces **140.sub.0-140.sub.n** using their respective PASID. Understand that in typical situations any one of execution circuits **130** may issue memory operations to any one of address spaces **140**. As used herein, the term “address

space” is used to refer to virtual address ranges accessible by a particular agent, such as a software agent. In different embodiments, supervisor software such as an operating system, virtual machine monitor, hypervisor or so forth may allocate different portions of a physical memory to different applications or other entities by creating a virtual address to physical address mapping in their address spaces. In addition, hardware circuitry of the processor, including a memory management unit, can be configured to provide address space protection such that only authorized agents are allowed to access a given address space. Understand while shown at this high level in the embodiment of FIG. 1, a data mover may include many other components and may be configured differently in other embodiments.

[0027] To enable a data mover's client to specify multiple address spaces in the job descriptor, the job descriptor may include additional fields for one or more PASID selectors for addresses specified in the descriptor. These PASID selectors specify additional PASIDs, called access PASIDs, that are to be accessed. Understand that these access PASIDs are in addition to a submitter PASID included in the job descriptor, which identifies the submitter's address space (i.e., submitter PASID). When the data mover receives a job descriptor having one or more PASID selectors, it accesses, via indexing using the PASID selector, the PASID permissions table entry to identify the access PASIDs and the submitter's permissions for the access PASIDs.

[0028] Referring now to FIG. 2, shown is a block diagram of a memory move job descriptor in accordance with an embodiment. In the embodiment of FIG. 2, job descriptor **200** may be an example job descriptor for performing a memory move operation in which data stored at a source location is copied to a destination location. With embodiments herein, job descriptor **200** enables this memory move operation to occur across multiple address spaces, thus reducing overhead of the operation. As shown, job descriptor **200** is a data structure including various fields. Of interest herein, a submitter PASID field **210** is used to identify a submitter of job descriptor **200**. Although the submitter PASID is part of the descriptor initialized by an unprivileged entity (e.g., application), the submitter PASID field cannot be initialized by unprivileged software. Submitter PASID field **210** is initialized by a CPU when sending the descriptor to the data mover or by a privileged entity (e.g., host device driver for the data mover) so the data mover always treats it as a trusted field. As shown, a completion record address field **215** may be used to identify an address within the address space of this submitter to which a completion record is to be written, upon completion of the memory move operation.

[0029] With embodiments herein, job descriptor **200** enables access to additional address spaces. More particularly here, in a first PASID selector field **220**, a source PASID selector may be provided to indicate a source access address space from which source data for the memory move operation may be obtained. Thus a source address field **225** may be used to identify a source address within this source address space from which data to be moved may be accessed. Similarly, in a second PASID selector field **230**, a destination PASID selector may be provided to indicate a destination access address space to which the source data is to be stored. Thus a destination address field **235** may be used to identify a destination address within this destination address space to which the data is to be copied. As further shown a destination notification address field **238** may be used to indicate where in the destination address space a notification regarding the completion can be written.

[0030] In the embodiment shown, note that job descriptor **200** may include other fields, including a completion interrupt handle field **250** to indicate where completion interrupt processing may occur. In some cases an additional PASID selector may identify an address space to which this field refers. Understand that additional fields, including a flags field **260** to store various flags, an operation field **270** to store an opcode (of the particular data move operation), among others also may be present. A transfer size field **280** may store size information for the move operation.

[0031] Additionally, while in the embodiment of FIG. 2, two PASID selector fields are shown, it is possible for a job descriptor to include more than two PASID selector fields, such as an additional

PASID selector field for purposes of interrupt handling, and one or more other PASID selector fields for other purposes such as reporting completions, errors, telemetry information or job notifications. Understand while this particular job descriptor is shown in FIG. 2, similar PASID selector fields may be included in other types of job descriptors as well.

[0032] Thus job descriptor **200** includes additional fields to support a copy across two different address spaces, and which may be submitted by a third application. As discussed above, the PASID permissions table (PPT) translates a PASID selector in a job descriptor to an access PASID and its access permissions. The PASID selector is used as an index into the PPT. In an embodiment, the PPT may include a plurality of entries, each containing the submitter PASID and the access PASID to be used for accessing the memory. The submitter PASID field in the PPT entry is validated against the job submitter's PASID. The PPT entry also contains the access permissions (e.g., read, write, and drain) allowed for the submitter. Each PPT entry also contains one or more address ranges in the access PASID's address space that the submitter is allowed to access. Each PPT entry may either enable or disable the address range check. If address range check is disabled, the entire address space of access PASID is accessible by the submitter. Each PPT entry may also optionally disable the submitter PASID check, if the owner desires to open an address window for global access by any submitter PASID.

[0033] In one embodiment, the PASID permissions table is located on the data mover device (in SRAM or on-device memory). In another embodiment, the PASID permissions table is located in system memory and is pointed to by device memory mapped input output (MMIO) registers (base PPT address and size). If implemented in memory, the data mover may cache frequently accessed PPT entries on the device in a caching structure to improve performance. The PPT may be implemented as single level table or multi-level table, in different implementations. One implementation of a multi-level table may be such that the first level holds information on the access PASID and submitter PASID pairs, and the set of next level tables hold information about different address windows and their corresponding permissions. An implementation may also allow the second level tables to be directly mapped into the address space of the owner PASID to allow direct modification of the address window and permissions parameters.

[0034] Referring now to FIG. 3, shown is a block diagram of a PASID permissions table according to one embodiment. As shown in FIG. 3, only one address range is implemented per PPT entry. Besides read and write access permissions, the PPT entry also specifies whether the Submitter is allowed to perform a drain operation on the access PASID, which allows software to ensure that all jobs which access the specified PASID have been completed in the device.

[0035] As shown in FIG. 3, PASID permissions table **300** includes a plurality of entries **310.sub.0-310_n**. As shown, an incoming PASID selector may be used to index into table **300** to access a given entry **310**. FIG. 3 further shows details of a representative entry within PASID permissions table **300** in accordance with an embodiment. As illustrated, example entry **310** includes various fields, including a submitter PASID field **311** to store a submitter PASID, which identifies a submitter of a given request (e.g., by way of work or job descriptor), and an access PASID field **312** to store an access PASID for another address space to be accessed. In addition to these fields, example entry **310** further includes a range size field **313** to indicate an address range, when address range permissions are implemented. Further in this regard, a range base address **314** identifies a base address of an address range, and range check enable **315**, which when set, indicates that address range permissions are enabled. As further shown, example entry **310** also includes additional permission fields, including a read permission field **316**, a write permission field **317** and a drain permission field **318**. Understand while shown with this particular implementation in the embodiment of FIG. 3, many variations and alternatives are possible.

[0036] The PASID permissions table can be set up by a privileged resource (e.g., a host driver for the data mover). To this end, applications may request the host driver to set up address ranges and access permissions. This may cause some amount of overhead if applications frequently want to

change address ranges and/or access permissions for those address ranges. For example, two MPI applications may want to share different address ranges with each other for short periods of time and this requires frequent changes to the PPT entry.

[0037] To reduce this overhead, embodiments may allow applications to directly change PPT entry information (e.g., address ranges and access permissions) without going through the privileged host driver. To this end, an application may send a job descriptor called a PPT setup descriptor for updating PPT entries. The PPT setup descriptor contains the PASID selector corresponding to the PPT entry to be modified. The PPT entry's access PASID field is validated against the descriptor Submitter's PASID before updating the PPT entry. In one embodiment, an application is allowed to change the following fields in the PPT entry through this descriptor; Range Size, Range Base Address, Range Check Enable, Read Permission, Write Permission, and Drain Permission. Note that the PPT setup descriptor is a special case where the descriptor submitter's PASID is validated against the access PASID field rather than the submitter PASID field. The PPT setup descriptor can only modify an existing valid PPT entry. It cannot allocate and setup a new PPT entry.

[0038] In some embodiments, before the PPT entry can be updated, the access PASID is drained to ensure previously submitted descriptors use the old PPT entry values. In one embodiment, it is software's responsibility to ensure that the access PASID is drained from the device. In another embodiment, it is the data mover's responsibility to ensure the access PASID is drained from the device as part of processing the PPT setup descriptor. In an implementation where the data mover caches frequently accessed PPT entries in a cache, it may also evict an entry from the cache before, during or after processing the PPT setup descriptor.

[0039] The data mover may handle incoming requests for multi-PASID secure access differently dependent on whether a peer application wants to access another peer application's address space, or whether a privileged resource (e.g., OS or VMM) wants to access its deprivileged application or VM's address space respectively.

[0040] When a VMM wants to access a VM's address space through the data mover, it requests the host driver to set up a PASID permissions table entry. The host driver allocates a PPT entry, sets it up and returns the corresponding PASID selector to the VMM. The VMM may setup the PPT entry to access the entire VM's address space (i.e., guest physical address (GPA) space) or part of the address space through an address range check in the PPT entry. Alternately, the VMM may allow the VM to implicitly or explicitly set the address range in the PPT entry to limit the VMM's access to VM's address space through the data mover. Such an operation may be used for vswitch and MKTME usages described above.

[0041] After this setup, the VMM can submit job descriptors to the data mover with the corresponding PASID selector. The data mover validates the submitter's access permissions for the access PASID and then accesses the VM's memory using the access PASID in the PPT entry.

[0042] For the case of an application accessing a peer application's address space, operation may proceed as follows. Assume Application A wants to share its address space with Application B. Application A first requests the data mover (e.g., via its host driver) to allow Application B access to its address space. Application A may initially provide an address range or the whole address space. The host driver allocates a PPT entry, sets it up, and returns the corresponding PASID selector (PPT entry index) to Application A. Application A then sends the PASID selector to Application B. This communication, in an embodiment, may be via an out-of-band mechanism (e.g., through inter-process communication). Application B may then use the PASID selector in job descriptors that it sends to the data mover.

[0043] If Application A seeks to change the sharing (either disable sharing or change the address range), it notifies Application B of the impending change. After receiving acknowledgement from Application B, Application A submits a PPT setup descriptor to the data mover to change the PPT entry. If address ranges were changed, after completion of the PPT setup descriptor, Application A communicates to Application B the new address ranges, to allow Application B to access those

address ranges.

[0044] Referring now to FIG. 4, shown is a flow diagram of a method in accordance with one embodiment. More specifically, method **400** of FIG. 4 is a method for setting up a privileged entity to access an address space associated with an unprivileged entity. As such, method **400** may be performed by a host driver, e.g., a privileged driver software that executes within a host OS. Accordingly, method **400** may be performed by hardware circuitry, firmware, software and/or combinations thereof.

[0045] As shown in FIG. 4, method **400** may be used to enable a VMM to access an address space of a VM that executes under the VMM. Of course many other examples of setting up address space permissions are possible, including allowing a privileged resource to access an address space associated with another privileged resource, a different type of unprivileged resource or so forth. Still further it is possible that unprivileged resources may seek to access address spaces associated with other unprivileged resources.

[0046] In FIG. 4, method **400** begins by receiving a request from a VMM (block **410**). More specifically, at block **410** this request received in the data mover driver may be for seeking access to an address space associated with a VM such as a guest application that executes under the VMM. In one embodiment, this request may be implemented as a setup request. As seen, in response to this request, at block **420** the data mover driver may allocate an entry in the PASID permissions table. Note that if no available entries are present, the request may be denied.

[0047] Next at block **430** this entry may be configured. More specifically, various information from the received request, e.g., setup request, may be stored in this entry. As an example, in response to the setup request, the data mover driver may initialize the first entry in the PASID permissions table to store a submitter PASID for the VMM to enable the VMM's secure access to the second address space, along with a PASID selector for the address space of the VM. In addition, various permissions to be granted to this submitter for the address space of the access PASID may be filled out. As examples, these permissions may include read and write access and drain access, among others. Of course other access permissions are possible in other embodiments. Then at block **440** a PASID selector is returned to the VMM. More specifically, this PASID selector may be used by the VMM to act as an index into the PASID permissions table when the VMM later issues a job descriptor to perform a memory operation, offloaded to the data mover, for the VMM as requester. It's to be understood while shown at this high level in the embodiment of FIG. 4, many variations and alternatives are possible.

[0048] Referring now to FIG. 5, shown is a flow diagram of a method in accordance with another embodiment. More specifically, method **500** of FIG. 5 is a method for setting up a unprivileged entity accesses, and updates to the same. As such, method **500** may be performed by a host driver and multiple, e.g., unprivileged applications. Accordingly, method **500** may be performed by hardware circuitry, firmware, software and/or combinations thereof.

[0049] In FIG. 5, method **500** begins by receiving a request from a first application to allow another application to access its address space, e.g., via a setup descriptor from a VMM (block **510**). In response to this request, at block **520** the data mover driver may allocate an entry in the PASID permissions table. Next at block **530** this entry may be configured. Then at block **540** a PASID selector is returned to the first application.

[0050] Next, at block **550** this PASID selector is sent to the application. In embodiments herein, the first application may send this PASID selector to the second application via an out-of-band mechanism, such as via an MPI interface. Thus at this point, with this provision of the PASID selector, the second application may issue a request to a data mover that seeks to access at least part of an address space associated with the first application.

[0051] In some cases, it is possible that the application may seek to change the permission and/or address range of the shared region at some point. Accordingly, with further reference to FIG. 5, it may be determined at diamond **555** whether there is a change to a sharing policy. For example, a

change in access permissions, and/or a permitted address range may occur. If it is determined that such change to sharing policy is to occur, control passes to block **560** where the first application may notify the second application regarding the change. In response to this notification, the second application may send an acknowledgment back to the first application, e.g., via the out-of-band mechanism. Thus as illustrated it is determined at diamond **565** whether an acknowledgment has been received.

[0052] When this acknowledgment is received such that the second application has been made aware of the impending change, control passes to block **570** where the first application may send a setup descriptor to the data mover. More specifically, the first application may send a PASID permissions table setup descriptor to request a change in permissions or address range for the second application. Once the data mover effects this change, e.g., by updating the appropriate entry of the PASID permissions table, it sends a completion back to the first application. And, as further illustrated in FIG. 5 when the first application receives this completion (as determined at diamond **580**), it communicates an updated address range and/or permissions to the second application (block **590**). As such, from this point going forward the second application may issue offload requests to the data mover only for addresses falling within the updated address range and/or updated permissions. Understand while shown at this high level in the embodiment of FIG. 5, many variations and alternatives are possible.

[0053] Referring now to FIG. 6, shown is a flow diagram of a method in accordance with yet another embodiment. As shown in FIG. 6, method **600** is a method for performing a data operation by a data mover on behalf of a client. Accordingly, method **600** may be performed by hardware circuitry, firmware, software and/or combinations thereof.

[0054] As illustrated, method **600** begins by receiving a job descriptor in the data mover from a first client (block **610**). Note that this client may be a software entity such as a privileged or non-privileged resource. In typical embodiments, this job descriptor may be received in the data mover by way of an offloading from a CPU or other processing circuit. Next, control passes to diamond **620** to determine whether the job descriptor includes one or more PASID selectors. If not, control passes to block **625** where the requested operation of the job descriptor may be performed in a single address space, namely the address space associated with this first client.

[0055] Otherwise when it is determined that the job descriptor includes at least one PASID selector, control passes to block **630**. At block **630**, this PASID selector may be used to index into the PASID permissions table. Thus based on this indexing, a given entry of the PASID permissions table may be accessed and information from this entry can be analyzed to determine whether the submitter PASID is validated (diamond **640**). Namely, this determination may be based on verifying that the submitter PASID received in the job descriptor matches the submitter PASID stored in the indexed entry. If not, control passes to block **650** where this job descriptor request may be aborted.

[0056] Assuming that the submitter PASID is validated, control next passes to diamond **660** to determine whether the address range of the job descriptor is permitted. For example, the job descriptor may identify source and destination addresses and at diamond **660**, it may be determined whether these addresses fall within an address range to which the first client is permitted access as per the permissions in the PPT entry. If not, control passes again to block **650** to abort the job descriptor request.

[0057] When it is determined that the job descriptor is thus validated (by way of validating submitter PASID, address ranges, and additional permissions (such as read/write permissions or so forth)), control next passes to block **670** where the operation of the job descriptor request may be performed. More specifically here, the operation may be performed securely across multiple address spaces, including possibly the address space of the first client, and one or more additional address spaces. To this end, optionally it may be determined whether there are additional PASID selectors in the job descriptor (diamond **665**). If so, control passes back to block **630** for further

analysis of an additional one or more entries within the PASID permissions table.

[0058] Finally, after the operation has been completed control passes to block **680**. At block **680** a completion may be communicated to the first client and also to a destination client. To this end, the data mover may send a completion message to the first client, namely to a completion record address identified in the job descriptor. In addition, the data mover may send another completion message to the destination client, and more specifically to a destination notification record address identified in the job descriptor. Understand while shown at this high level in the embodiment of FIG. **6**, many variations and alternatives are possible.

[0059] Referring now FIG. **7**, shown is a block diagram of a computing system in accordance with an embodiment of the present invention. As shown in FIG. **7**, computing system **700** may be any type of computing device, ranging from a small portable device such as smartphone, tablet computer, to larger devices such as client systems and even larger devices such as server systems or so forth. In the high level view of FIG. **7**, a system on chip (SoC) **710** couples to a system memory **720**, which may be implemented as a dynamic random access memory (DRAM). In addition, SoC **710** further couples to a smart controller **730** and a scalable IOV device **740**. As examples, smart controller **730** may be a smart NIC or other such device, while as one example scalable IOV device **740** may be implemented as a scalable storage device such as a non-volatile memory express (NVMe) device. Other examples of scalable IOV devices may include general purpose graphics processing unit (GPGPU) or other accelerators (crypto, compression, AI, etc.), or high speed network interface.

[0060] In the high level shown, SoC **700** includes one or more cores **712**, integrated memory controller **715** to interface with system memory **720** and an IO memory management unit (IOMMU) **718**. Of course many more components may be present within SoC **710**. In a virtualization environment, understand that a plurality of virtual machines (not shown for ease of illustration in FIG. **7**) may execute under a given virtual machine monitor or other supervisor software. Each of these entities may have independent address spaces mapped to portions of system memory **720**.

[0061] As further shown, smart controller **730** includes virtual devices (VDEVs) **732.sub.0,n** and, optionally a local memory **735**. In an embodiment, each VDEV **732** may be associated with a given virtual machine and may include one or more storages, such as registers to store associations for VMs and other information. To interface with scalable IOV device **740**, smart controller **730** may communicate to the scalable IOV device **740** by accessing its registers (similar to driver software running on CPU cores). As shown, scalable IOV device **740** itself includes Assignable Device Interfaces (ADIs) **742.sub.0,n**. With embodiments herein, each ADI **742** may be configured with multiple PASIDs, such as PASIDs **743.sub.0,1-.sub.0,n**. To this end, each ADI **742** may be associated with a given virtual machine and may include one or more storages, such as registers to store these PASIDs. Scalable IOV device **740** may further include at least one functional circuit **748**. In different examples, functional circuit **748** may implement a storage, an accelerator, or other device functionality that may be used by ADIs to perform the device's functions. And as shown, scalable IOV device **740** also may optionally include a local memory **745**.

[0062] Note that system **700** may take the form of a wide variety of computing platforms. In one particular example, computing system **700** may be implemented using one or more servers such as implemented rack-mounted system of a cloud services provider. In such use cases, a cloud services provider may use smart controller **730** (e.g., a smart NIC) to virtualize I/O device **740** (e.g., NVMe, GPU) to the VM. All upstream memory requests from smart controller **730** and scalable IOV device **740** go through IOMMU **718**, which translates the memory addresses from a guest physical address, e.g., of a corresponding VM, to a host physical address that is used to access the VM's data in system memory **720**. This translated host physical address is then routed to destination memory through the platform's physical address-based routing circuitry.

[0063] In an embodiment, VDEV emulation is done using a Virtual Device Composition Module

(VDCM) in the VMM or in a hardware component such as a smart controller. In some cases a smart NIC may be used to emulate VDEVs for the VMs. Typically, a VM's slow path I/O operations (such as control and configurations) are intercepted and emulated by the VDCM and fast path operations (e.g., work submissions) are submitted directly to IOV device **740**.

[0064] Smart controller **730** may intercept the fast-path work submission operations (and slow-path operations) from the VMs to perform some value-added services on the VM's I/O requests. After processing the work requests from the VMs, the smart controller creates new work descriptors in its own address space. Smart controller **730** in turn may submit the new I/O work descriptors to a corresponding ADI **742** of scalable IOV device **740**. Without an embodiment, if an ADI only supports a single PASID (i.e., can access only a single address space), undue overhead would be incurred. As an example, an ADI would have to access both work descriptors and data buffers from a single address space. This in turn would cause smart controller **730** to copy the data buffers from VM's address space to the smart controller's address space (since the new work descriptors are also created in the smart controller's address space) before submitting descriptors to the ADI, which would unduly impact performance overhead.

[0065] When a given ADI **742** of scalable IOV device **740** issues a direct memory access DMA request, it is tagged with a PASID. In turn, IOMMU **718** uses this PASID to identify the VM's address space and perform DMA remapping.

[0066] Instead with embodiments, scalable IOV device **740** may be optimized by enabling ADIs **742** to access multiple address spaces. For example, in one embodiment, ADIs **742** may access work descriptors from an address space of smart controller **730**, in addition to directly accessing data stored in a VM's address space.

[0067] As such, embodiments avoid incurring smart controller copy operations to copy I/O data from VM memory into its own memory, which incurs a large performance overhead. And further overhead reductions may be realized by enabling an ADI of a scalable IOV device to access both descriptors and data buffers stored in multiple address spaces, via configuration of ADIs with multiple PASIDs.

[0068] In an embodiment, such multiple PASIDs per scalable IOV ADI may improve performance, especially for smart controller usages. As an example, ADI **7420** can use one PASID (e.g., **743.sub.0,0**) to access work descriptors from one address space and another PASID (e.g., **743.sub.0,n**) to access data from another address space. With this arrangement, ADI **7420** can access multiple address spaces, and the host OS/VMM/smart controller can avoid copying data around to ensure all data is in one address space, thus reducing I/O virtualization overhead.

[0069] In embodiments, scalable I/O devices support a work submission interface using a work queue in system memory. The work queue contains work descriptors, which describe the I/O operation to be performed. Note the terms "work descriptor" and "job descriptor" may be used interchangeably herein. Each I/O work descriptor may contain one or more addresses pointing to data. I/O requests may be submitted by enqueueing work descriptors in the work or descriptor queue and then informing the I/O device. In an embodiment this notification may be implemented using a doorbell mechanism (typically a MMIO register in the device). The device first reads the descriptor from the queue and processes it. As part of the processing, the device also reads/writes the addresses in the descriptor to access the data and perform the requested function.

[0070] Referring now to FIG. **8**, shown is a block diagram of a system in accordance with another embodiment. As shown in FIG. **8**, system **800**, which may be implemented similarly to system **700** of FIG. **7**, shows interaction between software entities, including virtual machines or containers **810.sub.0,n** and corresponding smart controller **830** and scalable IOV device **840**. As further shown, interaction between these components and an underlying host software, e.g. a VMM/host OS **820**, also is illustrated.

[0071] As shown, each VM **810** may include corresponding descriptor queues **812** and data buffers **814**. Smart controller **830**, via VDEVs **832.sub.0,n**, may generate one or more descriptor queues

825 stored in host memory **820** (in smart controller's address space). In turn, by way of multiple PASIDs **843** (namely **843.sub.0,0-0,1;n,0-n,1**), ADIs **842** may access multiple address spaces, including a corresponding descriptor queue **825** in host memory **820** and a data buffer **814** in corresponding VM memory **810**. In this way, at least one functional circuit **848** within scalable IOV device **840** may be shared by ADIs **842** to process the work descriptors (or in other words, perform the device's functions).

[0072] In the embodiment of FIG. 8, an ADI **842** may be assigned to a given VM by emulating a VDEV **832** in smart controller **830**. Note in another embodiment, the VDEV can be emulated in the VMM/Host OS using a VDCM.

[0073] A guest driver in VM **810.sub.0** prepares descriptor queue **812.sub.0**, which contains I/O work descriptors. These I/O work descriptors are submitted to VDEV **832** using a device specific method (e.g., a doorbell). This triggers smart controller **830** to fetch the I/O descriptors from VM's descriptor queue **812.sub.0**. After processing the descriptors, smart controller **830** prepares new I/O descriptors in a descriptor queue **825** in host memory **820**. Note in another embodiment, a smart controller may prepare new I/O descriptors in a descriptor queue in controller's local memory.

[0074] As shown in FIG. 8, ADI **843.sub.0** is configured to point to descriptor queue **825** of smart controller **830** via a descriptor PASID **843.sub.0,1**, which is used to access descriptor queue **825** in host memory **820**. In turn ADI **843.sub.0** may access data buffer **814.sub.0** in VM memory **810.sub.0** using a data PASID **843.sub.0,1**.

[0075] After preparing the I/O descriptors in host descriptor queue **825**, smart controller **830** submits them to ADIs **842** using a device specific method (e.g., doorbell). This triggers ADIs **842** to fetch the I/O work descriptor from host memory **820** using descriptor PASIDs **843**. As part of processing the descriptors, ADIs **842** may access data stored in data buffers **814** of VM memory **810** using data PASIDs **843**. This allows the VMM/smart controller to avoid copying the data buffers from VM memory to host memory beforehand, improving I/O virtualization performance.

[0076] In an embodiment in which a smart controller creates a descriptor queue in its own local memory, descriptor fetch from the ADI is routed to the local memory. To this end, a VMM may enable peer-to-peer DMA from scalable IOV device to smart controller. On ADI's descriptor fetch, an IOMMU performs an address translation to obtain a physical address of the smart controller's local descriptor queue. The platform's physical address-based routing ensures that the descriptor queue access is redirected to the smart controller's memory for a peer-to-peer DMA.

[0077] In another embodiment, an ADI may support additional PASIDs (i.e., more than 2 PASIDs) to access other device-specific data from additional address spaces. For example, a scalable IOV device may store message signaled interrupt (MSI) interrupt messages (i.e., MSI address and data) in a table (e.g., an interrupt table) in host memory. The interrupt table may be created in a separate address space than descriptor queues and data buffers. Consequently, the ADIs may use a third PASID to fetch the MSI message from the interrupt table for generating an interrupt to notify (e.g., requester and/or target) as to completion of descriptors.

[0078] Referring now to FIG. 9, shown is a flow diagram of a method in accordance with yet another embodiment of the present invention. More specifically, method **900** is a method for accessing multiple address spaces by a scalable IOV device in accordance with an embodiment. Accordingly, method **900** may be performed by hardware circuitry, firmware, software and/or combinations thereof.

[0079] As illustrated, method **900** begins by preparing a first I/O job descriptor for a first application (block **910**). In one embodiment, a guest driver, e.g., of a given VM may prepare this job descriptor. Next at block **920** this first I/O job descriptor may be submitted to a virtual device emulated by a smart controller. In an embodiment, a device specific method such as a doorbell mechanism may be used to communicate this I/O job descriptor.

[0080] Still with reference to FIG. 9, next at block **930** the smart controller may process this first I/O job descriptor to prepare a second I/O job descriptor and store it in a descriptor queue in host

memory. Note that this descriptor queue may be associated with a corresponding ADI, namely an ADI associated with the first application. Then, the smart controller may submit this job to the scalable I/O virtualization device (block **940**). More particularly, this second I/O job descriptor may be submitted to an ADI of the scalable I/O virtualization device. In response to this submission, at block **950** the ADI may obtain the job descriptor from the descriptor queue in host memory. To this end, the ADI may use a descriptor PASID to obtain the second I/O job descriptor. Then at block **960** the ADI may process this second I/O job descriptor to determine the requested operation, permissions and other information. Then at block **970** in response to this second I/O job descriptor, data within an address space of the first application may be accessed. More specifically, the ADI may use a data PASID that points to an address space, e.g., of a VM including a data buffer that includes the data to be accessed. Understand while shown at this high level in the embodiment of FIG. **9**, many variations and alternatives are possible.

[0081] The following examples pertain to further embodiments.

[0082] In one example, an apparatus comprises an IOV device including: at least one function circuit to be shared by a plurality of VMs; and a plurality of ADIs coupled to the at least one function circuit, where each of the plurality of ADIs is to be associated with one of the plurality of VMs and comprises a first PASID field to store a first PASID to identify a descriptor queue stored in a host address space and a second PASID field to store a second PASID to identify a data buffer located in a VM address space.

[0083] In an example, in response to a request from a first VM, a first ADI is to fetch a first IO work descriptor from the descriptor queue stored in the host address space using the first PASID and in response to the first IO work descriptor, to access a first data buffer of a first VM address space using the second PASID.

[0084] In an example, the first VM is to issue the request comprising a second IO work descriptor to a virtual device coupled to the IOV device.

[0085] In an example, a first virtual device emulated by a smart controller, in response to the request, is to access the second IO work descriptor from a second descriptor queue of the first VM address space, process the second IO work descriptor, prepare the first IO work descriptor from the second IO work descriptor and store the first IO work descriptor in the descriptor queue stored in the smart controller or the host address space.

[0086] In an example, the smart controller is to issue a message to the first ADI to cause the first ADI to fetch the first IO work descriptor from the descriptor queue stored in the smart controller or the host address space using the first PASID.

[0087] In an example, in response to the first IO work descriptor, the first ADI is to access the first data buffer located in the first VM address space using the second PASID.

[0088] In an example, the smart controller is to issue the message to the first ADI via a doorbell mechanism.

[0089] In an example, the apparatus further comprises a SoC coupled to the IOV device, where the SoC comprises a I/O memory management unit to receive a first virtual address from the first ADI and send a first physical address corresponding to the first virtual address to the first VM address space to enable the access to a first data buffer located in the first VM address space.

[0090] In an example, the SoC is coupled with one or more of the smart controller and the IOV device.

[0091] In an example, at least some of the plurality of ADIs comprise a third PASID field to store a third PASID to identify another address space to which the ADI is allowed access.

[0092] In an example, a first ADI of the at least some of the plurality of ADIs is to access an interrupt table stored in the another address space using the third PASID.

[0093] In another example, a method comprises: fetching, by a first ADI of an IOV device, a first job descriptor from a descriptor queue of a second address space, using a descriptor PASID of the first ADI, where the descriptor PASID is associated with the second address space; and in response

to the first job descriptor, accessing data stored in a first data buffer of a first address space associated with a first entity, using a data PASID of the first ADI, where the data PASID is associated with the first address space that contains the first data buffer.

[0094] In an example, the method further comprises fetching the first job descriptor in response to an indication received from a first virtual device of a smart controller coupled to the IOV device.

[0095] In an example, the method further comprises fetching, by the first virtual device, a second job descriptor from a descriptor queue of the first address space, in response to an indication received from the first entity.

[0096] In an example, the method further comprises: processing the second job descriptor, generating the first job descriptor from the second job descriptor, and storing the first job descriptor into the descriptor queue of the second address space comprising a host memory; and after storing the first job descriptor into the descriptor queue of the host memory, indicating to the first ADI presence of the first job descriptor in the descriptor queue of the host memory.

[0097] In an example, the method further comprises after processing the data, writing, by the first ADI, an interrupt message into an interrupt table stored in a third address space, using a third PASID of the first ADI that points to the interrupt table, where the interrupt message comprises a completion for an operation on the data.

[0098] In an example, the method further comprises configuring the first ADI with the descriptor PASID and the data PASID.

[0099] In another example, a computer readable medium including instructions is to perform the method of any of the above examples.

[0100] In a further example, a computer readable medium including data is to be used by at least one machine to fabricate at least one integrated circuit to perform the method of any one of the above examples.

[0101] In a still further example, an apparatus comprises means for performing the method of any one of the above examples.

[0102] In another example, a computing system comprises: a processor having one or more cores and a memory controller to interface with a system memory; a smart controller coupled to the processor, where the smart controller is to emulate a first virtual device, and in response to a request from a first VM access a first work descriptor from a first descriptor queue of a first address space of the first VM, prepare a second work descriptor from the first work descriptor and store the second work descriptor in a descriptor queue of a host address space; and an IOV device coupled to the smart controller. The IOV device may include: at least one function circuit to be shared by a plurality VMs; and a plurality of ADIs coupled to the at least one function circuit, where each of the plurality of ADIs is to be associated with one of the plurality of VMs and comprises a first PASID field to store a first PASID to identify the descriptor queue of the host address space and a second PASID field to store a second PASID to identify a data buffer located in an address space of one of the plurality of VMs. The computing system may further include the system memory coupled to the processor, where the system memory is to be used as a backing memory for the host address space and address spaces of the plurality of VMs.

[0103] In an example, in response to the request from the first VM, a first ADI is to fetch the second work descriptor from the descriptor queue of the host address space using the first PASID and in response to the second work descriptor, to access a first data buffer of the address space of the first VM using the second PASID.

[0104] In an example, in response to the request from the first VM to the first virtual device, the smart controller is to issue another request to the first ADI to cause the first ADI to fetch the second work descriptor from the descriptor queue of the host address space.

[0105] Understand that various combinations of the above examples are possible.

[0106] Note that the terms “circuit” and “circuitry” are used interchangeably herein. As used herein, these terms and the term “logic” are used to refer to alone or in any combination, analog

circuitry, digital circuitry, hard wired circuitry, programmable circuitry, processor circuitry, microcontroller circuitry, hardware logic circuitry, state machine circuitry and/or any other type of physical hardware component. Embodiments may be used in many different types of systems. For example, in one embodiment a communication device can be arranged to perform the various methods and techniques described herein. Of course, the scope of the present invention is not limited to a communication device, and instead other embodiments can be directed to other types of apparatus for processing instructions, or one or more machine readable media including instructions that in response to being executed on a computing device, cause the device to carry out one or more of the methods and techniques described herein.

[0107] Embodiments may be implemented in code and may be stored on a non-transitory storage medium having stored thereon instructions which can be used to program a system to perform the instructions. Embodiments also may be implemented in data and may be stored on a non-transitory storage medium, which if used by at least one machine, causes the at least one machine to fabricate at least one integrated circuit to perform one or more operations. Still further embodiments may be implemented in a computer readable storage medium including information that, when manufactured into a SoC or other processor, is to configure the SoC or other processor to perform one or more operations. The storage medium may include, but is not limited to, any type of disk including floppy disks, optical disks, solid state drives (SSDs), compact disk read-only memories (CD-ROMs), compact disk rewritables (CD-RWs), and magneto-optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic random access memories (DRAMs), static random access memories (SRAMs), erasable programmable read-only memories (EPROMs), flash memories, electrically erasable programmable read-only memories (EEPROMs), magnetic or optical cards, or any other type of media suitable for storing electronic instructions.

[0108] While the present invention has been described with respect to a limited number of embodiments, those skilled in the art will appreciate numerous modifications and variations therefrom. It is intended that the appended claims cover all such modifications and variations as fall within the true spirit and scope of this present invention.

Claims

1. An apparatus, comprising: one or more cores to execute instructions; a memory controller to couple the one or more cores to a system memory; and a data mover device coupled to the memory controller to perform a copy operation indicated by a descriptor stored in a descriptor queue in a first address space, the copy operation to copy data from a source address space corresponding to a source process address space identifier (PASID) associated with a first virtual machine (VM) to a destination address space corresponding to a destination PASID associated with a second VM, the data mover device comprising: a plurality of assignable device interfaces (ADIs), including a first ADI associated with the first VM and a second ADI associated with the second VM; and circuitry to fetch the descriptor from the descriptor queue, the descriptor to indicate the copy operation, a source data buffer in the source address space, and a destination data buffer in the destination address space, wherein the data mover device is to access the data from the source data buffer in the source address space using the first PASID and is to copy the data to the destination data buffer in the destination address space using the second PASID based on the descriptor fetched from the descriptor queue.
2. The apparatus of claim 1, wherein the descriptor includes one or more fields to indicate a corresponding one or more permissions tables from which to determine the first PASID and the second PASID.
3. The apparatus of claim 1, wherein the data mover device is to use the first ADI to access the data from the source data buffer and is to use the second ADI to store the data in the destination data

buffer.

4. The apparatus of claim 1, wherein the first address space is associated with a third PASID, the circuitry to fetch the descriptor with the third PASID.

5. The apparatus of claim 1, wherein the descriptor comprises a first Input/output (IO) work descriptor and the descriptor queue comprises a first descriptor queue, wherein a first virtual device provided by a smart controller, in response to a request, is to access a second IO work descriptor from a second descriptor queue, process the second IO work descriptor, prepare the first IO work descriptor from the second IO work descriptor, and store the first IO work descriptor in the first descriptor queue.

6. The apparatus of claim 1, further comprising an input/output (IO) memory management unit (IOMMU) to receive a first virtual address from the first ADI including the first PASID and send a first physical address corresponding to the first virtual address to the first ADI to enable the access to the first data buffer.

7. The apparatus of claim 1, wherein the first ADI is to access an interrupt table stored in the address space accessible to the second VM using a third PASID.

8. A method comprising: executing, by one or more cores of an apparatus, instructions; coupling, by a memory controller of the apparatus, the one or more cores to a system memory; performing, by a data mover device coupled to the memory controller, a copy operation indicated by a descriptor stored in a descriptor queue in a first address space, the copy operation to copy data from a source address space corresponding to a source process address space identifier (PASID) associated with a first virtual machine (VM) to a destination address space corresponding to a destination PASID associated with a second VM, wherein the data mover device comprises a plurality of assignable device interfaces (ADIs), including a first ADI associated with the first VM and a second ADI associated with the second VM, and wherein the copy operation comprises: fetching the descriptor from the descriptor queue, the descriptor to indicate the copy operation, a source data buffer in the source address space, and a destination data buffer in the destination address space, wherein the data mover device is to access the data from the source data buffer in the source address space using the first PASID and is to copy the data to the destination data buffer in the destination address space using the second PASID based on the descriptor fetched from the descriptor queue.

9. The method of claim 8, wherein the descriptor includes one or more fields to indicate a corresponding one or more permissions tables from which to determine the first PASID and the second PASID.

10. The method of claim 8, wherein the data mover device is to use the first ADI to access the data from the source data buffer and is to use the second ADI to store the data in the destination data buffer.

11. The method of claim 8, wherein the first address space is associated with a third PASID, the descriptor is fetched with the third PASID.

12. The method of claim 8, wherein the descriptor comprises a first Input/output (IO) work descriptor and the descriptor queue comprises a first descriptor queue, wherein a first virtual device provided by a smart controller, in response to a request, is to access a second IO work descriptor from a second descriptor queue, process the second IO work descriptor, prepare the first IO work descriptor from the second IO work descriptor, and store the first IO work descriptor in the first descriptor queue.

13. The method of claim 8, further comprising an input/output (IO) memory management unit (IOMMU) to receive a first virtual address from the first ADI including the first PASID and send a first physical address corresponding to the first virtual address to the first ADI to enable the access to the first data buffer.

14. The method of claim 8, wherein the first ADI is to access an interrupt table stored in the address space accessible to the second VM using a third PASID.

15. A non-transitory machine-readable storage medium storing instructions, which when executed by an apparatus, cause the apparatus to perform: executing, by one or more cores of the apparatus, instructions; coupling, by a memory controller of the apparatus, the one or more cores to a system memory; and performing, by a data mover device coupled to the memory controller, a copy operation indicated by a descriptor stored in a descriptor queue in a first address space, the copy operation to copy data from a source address space corresponding to a source process address space identifier (PASID) associated with a first virtual machine (VM) to a destination address space corresponding to a destination PASID associated with a second VM, wherein the data mover device comprises a plurality of assignable device interfaces (ADIs), including a first ADI associated with the first VM and a second ADI associated with the second VM, and wherein the copy operation comprises: fetching the descriptor from the descriptor queue, the descriptor to indicate the copy operation, a source data buffer in the source address space, and a destination data buffer in the destination address space, wherein the data mover device is to access the data from the source data buffer in the source address space using the first PASID and is to copy the data to the destination data buffer in the destination address space using the second PASID based on the descriptor fetched from the descriptor queue.

16. The non-transitory machine-readable storage medium of claim 15, wherein the descriptor includes one or more fields to indicate a corresponding one or more permissions tables from which to determine the first PASID and the second PASID.

17. The non-transitory machine-readable storage medium of claim 15, wherein the data mover device is to use the first ADI to access the data from the source data buffer and is to use the second ADI to store the data in the destination data buffer.

18. The non-transitory machine-readable storage medium of claim 15, wherein the first address space is associated with a third PASID, the descriptor is fetched with the third PASID.

19. The non-transitory machine-readable storage medium of claim 15, wherein the descriptor comprises a first Input/output (IO) work descriptor and the descriptor queue comprises a first descriptor queue, wherein a first virtual device provided by a smart controller, in response to a request, is to access a second IO work descriptor from a second descriptor queue, process the second IO work descriptor, prepare the first IO work descriptor from the second IO work descriptor, and store the first IO work descriptor in the first descriptor queue.

20. The non-transitory machine-readable storage medium of claim 15, further comprising an input/output (IO) memory management unit (IOMMU) to receive a first virtual address from the first ADI including the first PASID and send a first physical address corresponding to the first virtual address to the first ADI to enable the access to the first data buffer.
